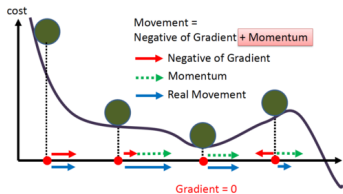
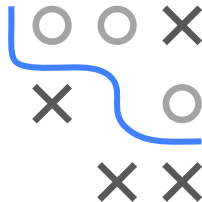


# Optimization in Machine Learning

## Advanced first order methods GD with Momentum



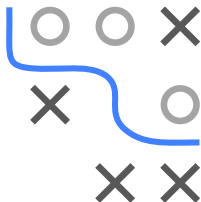
### Learning goals

- Recap of GD problems
- Momentum definition
- Unrolling formula
- Examples
- Nesterov

## RECAP: WEAKNESSES OF GRADIENT DESCENT

- **Zig-zagging behavior:** For ill-conditioned problems, GD moves with a zig-zag course to the optimum, since the gradient points approximately orthogonal in the shortest direction to the minimum.
- **Slow crawling:** Gradient may vanish rapidly close to stationary points (e.g. saddle points) and hence also slows down progress.
- **Trapped in stationary points:** In some functions, GD converges to stationary points (e.g. saddle points) since gradient on all sides is fairly flat and the step size is too small to pass this flat part.

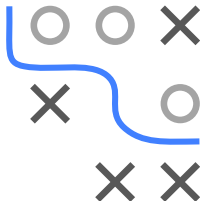
**Aim:** More efficient algorithms which quickly reach the minimum.





# GD WITH MOMENTUM: CHARACTERISTICS

- Length of a single step depends on how large and aligned a sequence of gradients is
- Length of a single step grows if many successive gradients point in the same direction
- $\varphi$  determines how strongly previous gradients are included in  $\nu$
- Common values for  $\varphi$  are 0.5, 0.9 and even 0.99
- In general, the larger  $\varphi$  is in relation to  $\alpha$ , the more strongly previous gradients influence the current direction
- **Special case**  $\varphi = 0$ : “vanilla” gradient descent
- **Intuition:** GD with “short term memory” for the direction of motion



## MOMENTUM: ANALYSIS

$$\boldsymbol{\nu}^{[1]} = \varphi \boldsymbol{\nu}^{[0]} - \alpha \nabla f(\mathbf{x}^{[0]})$$

$$\mathbf{x}^{[1]} = \mathbf{x}^{[0]} + \varphi \mathbf{v}^{[0]} - \alpha \nabla f(\mathbf{x}^{[0]})$$

$$\begin{aligned}\mathbf{v}^{[2]} &= \varphi \mathbf{v}^{[1]} - \alpha \nabla f(\mathbf{x}^{[1]}) \\ &= \varphi(\varphi \mathbf{v}^{[0]} - \alpha \nabla f(\mathbf{x}^{[0]})) - \alpha \nabla f(\mathbf{x}^{[1]})\end{aligned}$$

$$\mathbf{x}^{[2]} = \mathbf{x}^{[1]} + \varphi(\varphi \mathbf{v}^{[0]} - \alpha \nabla f(\mathbf{x}^{[0]})) - \alpha \nabla f(\mathbf{x}^{[1]})$$

$$\begin{aligned}\mathbf{v}^{[3]} &= \varphi \mathbf{v}^{[2]} - \alpha \nabla f(\mathbf{x}^{[2]}) \\ &= \varphi(\varphi(\varphi \mathbf{v}^{[0]} - \alpha \nabla f(\mathbf{x}^{[0]})) - \alpha \nabla f(\mathbf{x}^{[1]})) - \alpha \nabla f(\mathbf{x}^{[2]})\end{aligned}$$

$$\mathbf{x}^{[3]} = \mathbf{x}^{[2]} + \textcolor{teal}{\varphi}(\textcolor{teal}{\varphi}(\textcolor{teal}{\varphi}\boldsymbol{\nu}^{[0]} - \alpha\nabla f(\mathbf{x}^{[0]})) - \alpha\nabla f(\mathbf{x}^{[1]})) - \alpha\nabla f(\mathbf{x}^{[2]})$$

$$= \mathbf{x}^{[2]} + \varphi^3 \boldsymbol{\nu}^{[0]} - \varphi^2 \alpha \nabla f(\mathbf{x}^{[0]}) - \varphi \alpha \nabla f(\mathbf{x}^{[1]}) - \alpha \nabla f(\mathbf{x}^{[2]})$$

$$= \mathbf{x}^{[2]} - \alpha(\varphi^2 \nabla f(\mathbf{x}^{[0]}) + \varphi^1 \nabla f(\mathbf{x}^{[1]}) + \varphi^0 \nabla f(\mathbf{x}^{[2]})) + \varphi^3 \boldsymbol{\nu}^{[0]}$$

$$\mathbf{x}^{[t+1]} = \mathbf{x}^{[t]} - \alpha \sum_{j=0}^t \varphi^j \nabla f(\mathbf{x}^{[t-j]}) + \varphi^{t+1} \boldsymbol{\nu}^{[0]}$$

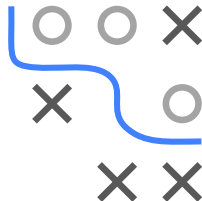
# MOMENTUM: INTUITION

Suppose momentum always observes the same gradient  $\nabla f(\mathbf{x}^{[t]})$ :

$$\begin{aligned}\mathbf{x}^{[t+1]} &= \mathbf{x}^{[t]} - \alpha \sum_{j=0}^t \varphi^j \nabla f(\mathbf{x}^{[t-j]}) + \varphi^{t+1} \boldsymbol{\nu}^{[0]} \\ &= \mathbf{x}^{[t]} - \alpha \nabla f(\mathbf{x}^{[t]}) \sum_{j=0}^t \varphi^j + \varphi^{t+1} \boldsymbol{\nu}^{[0]} \\ &= \mathbf{x}^{[t]} - \alpha \nabla f(\mathbf{x}^{[t]}) \frac{1 - \varphi^{t+1}}{1 - \varphi} + \varphi^{t+1} \boldsymbol{\nu}^{[0]} \\ &\rightarrow \mathbf{x}^{[t]} - \alpha \nabla f(\mathbf{x}^{[t]}) \frac{1}{1 - \varphi} \quad \text{for } t \rightarrow \infty.\end{aligned}$$

Momentum accelerates along  $-\nabla f(\mathbf{x}^{[t]})$  to terminal velocity yielding step size  $\alpha/(1 - \varphi)$ .

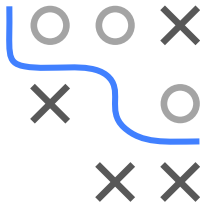
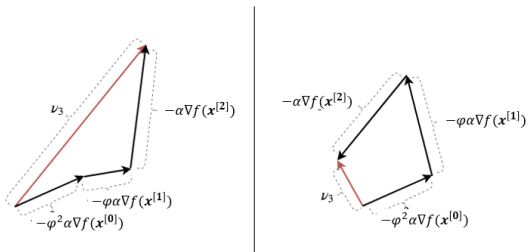
**Example:** Momentum with  $\varphi = 0.9$  corresponds to a tenfold increase in original step size  $\alpha$  compared to vanilla gradient descent



# MOMENTUM: INTUITION (EXAMPLE)

Vector  $\nu^{[3]}$  (for  $\nu^{[0]} = 0$ ):

$$\begin{aligned}\nu^{[3]} &= \varphi(\varphi(\varphi\nu^{[0]} - \alpha\nabla f(\mathbf{x}^{[0]})) - \alpha\nabla f(\mathbf{x}^{[1]})) - \alpha\nabla f(\mathbf{x}^{[2]}) \\ &= -\varphi^2\alpha\nabla f(\mathbf{x}^{[0]}) - \varphi\alpha\nabla f(\mathbf{x}^{[1]}) - \alpha\nabla f(\mathbf{x}^{[2]})\end{aligned}$$



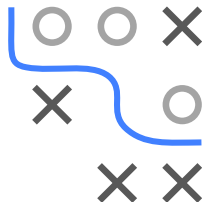
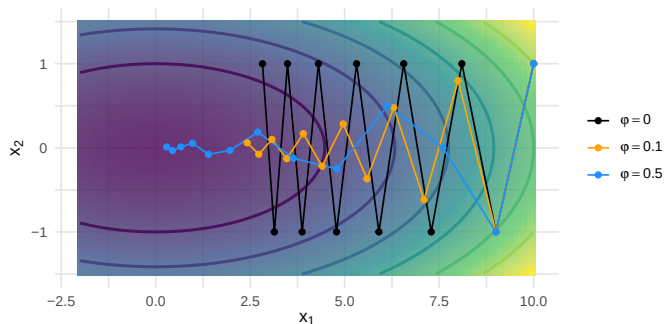
Successive gradients pointing in same/different directions increase/decrease velocity.

Further geometric intuitions and detailed explanations: [► Goh 2017](#)

# GD WITH MOMENTUM: ZIG-ZAG BEHAVIOR

Consider a two-dimensional quadratic form  $f(\mathbf{x}) = x_1^2/2 + 10x_2^2$ .

Let  $\mathbf{x}^{[0]} = (10, 1)^\top$  and  $\alpha = 0.1$ .

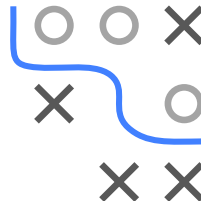
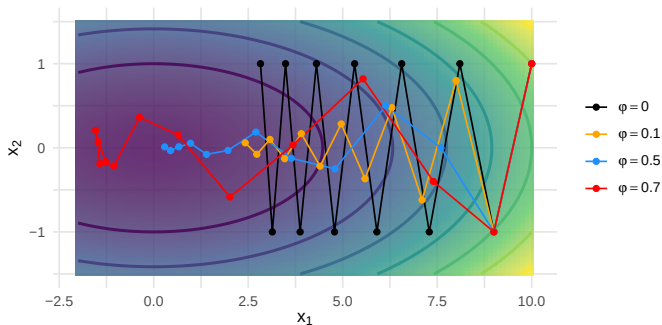


Momentum reduces zig-zagging and accelerates convergence in the direction of the minimum compared to vanilla GD.

# MOMENTUM: CAUTION

## Caution:

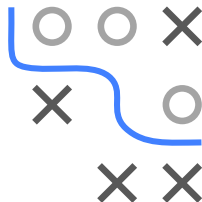
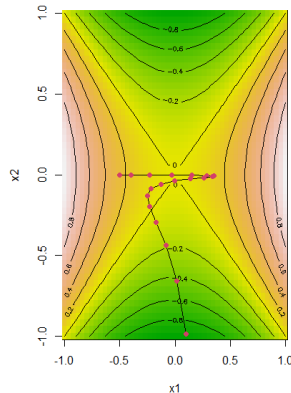
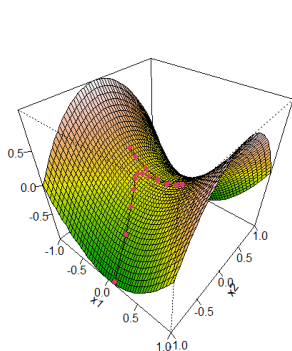
- If momentum is too high, minimum is possibly missed
- We might go back and forth around or between local minima



# GD WITH MOMENTUM: SADDLE POINTS

Consider the two-dimensional quadratic form  $f(\mathbf{x}) = x_1^2 - x_2^2$  with a saddle point at  $(0, 0)^\top$ .

Let  $\mathbf{x}^{[0]} = (-1/2, 10^{-3})^\top$  and  $\alpha = 0.1$ .



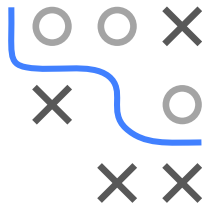
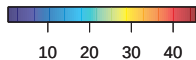
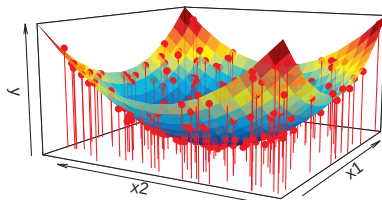
GD was slowing down at the saddle point (vanishing gradient); GD with momentum “breaks out” of the saddle point and moves on.

# ERM FOR NN WITH GD

Let  $\mathcal{D} = ((\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)}))$ , with  $y = x_1^2 + x_2^2$  and minimize

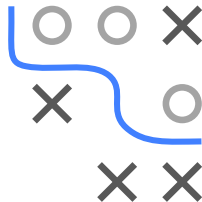
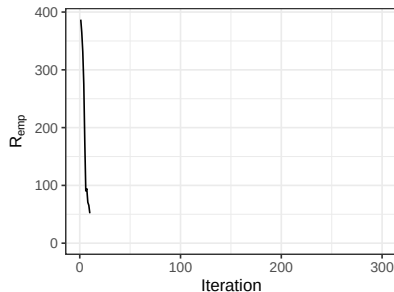
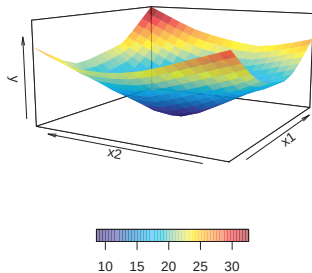
$$\mathcal{R}_{\text{emp}}(\boldsymbol{\theta}) = \sum_{i=1}^n (f(\mathbf{x} \mid \boldsymbol{\theta}) - y^{(i)})^2$$

where  $f(\mathbf{x} \mid \boldsymbol{\theta})$  is a neural network with two hidden layers (16 ReLU units each).



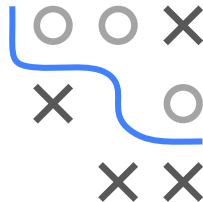
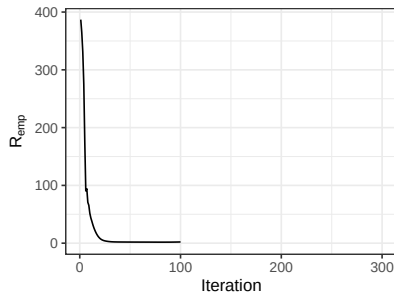
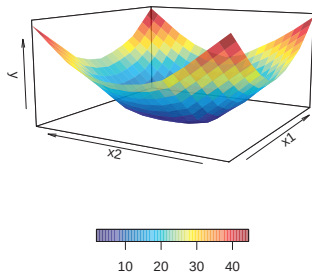
# ERM FOR NN WITH GD: 10 ITERATIONS

After 10 iters of GD with momentum  $\varphi = 0.5$ :



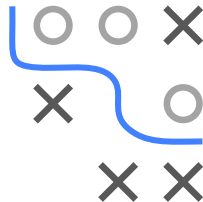
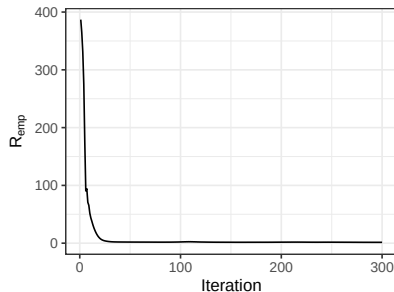
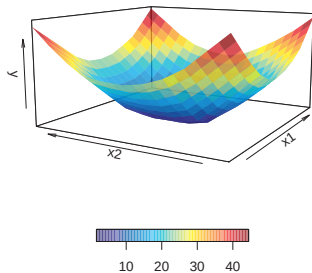
# ERM FOR NN WITH GD: 100 ITERATIONS

After 100 iters of GD with momentum  $\varphi = 0.5$ :



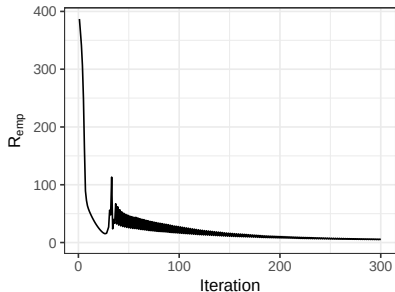
# ERM FOR NN WITH GD: 300 ITERATIONS

After 300 iters of GD with momentum  $\varphi = 0.5$ :

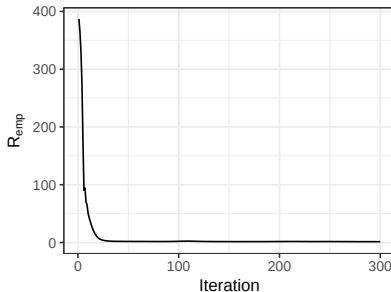


# GRADIENT DESCENT WITH AND WITHOUT MOMENTUM

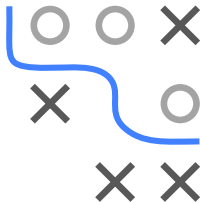
Without momentum ( $\varphi = 0$ ):



With momentum ( $\varphi = 0.5$ ):



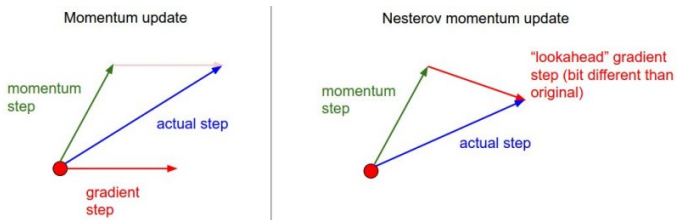
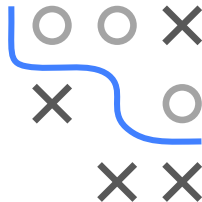
Momentum eliminates the oscillations of vanilla GD (potentially caused by overshooting in high-curvature directions with a too-large learning rate), resulting in faster, smoother convergence



# NESTEROV ACCELERATED GRADIENT

- Slightly modified version: **Nesterov accelerated gradient**
- Stronger theoretical convergence guarantees for convex functions
- Avoid moving back and forth near optima

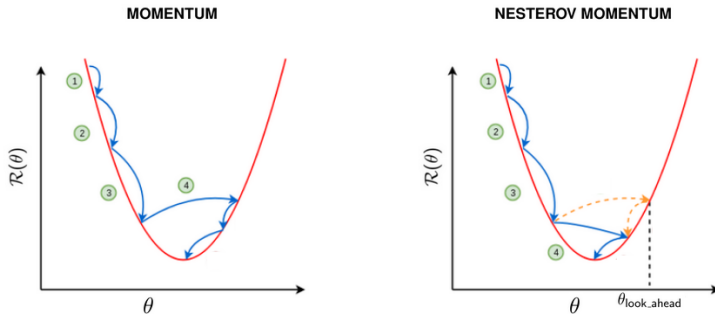
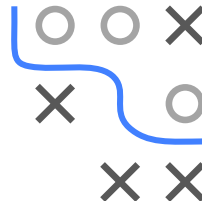
$$\begin{aligned}\nu^{[t+1]} &= \varphi \nu^{[t]} - \alpha \nabla f(\mathbf{x}^{[t]} + \varphi \nu^{[t]}) \\ \mathbf{x}^{[t+1]} &= \mathbf{x}^{[t]} + \nu^{[t+1]}\end{aligned}$$



► [Click for source](#)

Nesterov momentum update evaluates gradient at the “look-ahead” position.

# MOMENTUM VS. NESTEROV



► [Click for source](#)

GD with momentum (**left**) vs. GD with Nesterov momentum (**right**).  
Near minima, momentum makes a large step due to gradient history.  
Nesterov mom. “looks ahead” and reduces effect of gradient history.